

A Voyage to the Kernel



Part 12

Segment 3.1, Day 11

*We*re entering a new phase in our journey—kernel programming. In the first part, we'll cover a broad introduction to the Linux platform for newbies, along with some history.

The Linux platform

The UNIX platform showed us the power of the multi-layer security architecture, and the beauty of a well-organised structure in terms of subsystems and layers. This is perhaps one aspect that inspired Linus Torvalds, who was then studying computer science at the University of Helsinki, to develop a free operating system. (It should be noted that there is still an ongoing project to remove the non-free portion of the kernel code.)

In 1990 he wrote:

From: torvalds@klaava.Helsinki.FI (Linus Benedict Torvalds)

News:groups: comp.os.minix

Subject: Gcc-1.40 and a posix-question

Message-ID: <1991Jul3.100050.9886@klaava.Helsinki.FI>

Date: 3 Jul 91 10:00:50 GMT

Hello netlanders,

Due to a project I'm working on (in minix), I'm interested in the posix standard definition. Could somebody please point me to a (preferably) machine-readable format of the latest posix rules? Ptp-sites would be nice.

The 90s home PCs were powerful enough to run a full-blown UNIX OS. So Linus thought it would be a good idea to come out with a freely available academic version of UNIX. He based his project on Minix. To quote him again:

Date: Aug 26 1991, 11:12 am

Subject: What would you like to see most in minix?

To: comp.os.minix

Hello everybody out there using minix -

I'm doing a (free) operating system (just a hobby, won't be big and professional like gru) for 386(486) AT clones. This has been brewing since april, and is starting to get ready. I'd like any feedback on things people like/dislike in minix, as my OS resembles it somewhat (same physical layout of the file-system (due to practical reasons) among other things).

I've currently ported bash(1.08) and gcc(1.40), and things seem to work. This implies that I'll get something practical within a few months, and I'd like to know what features most people would want. Any suggestions are welcome, but I won't promise I'll implement them :-)

Linus (torva...@kruuna.helsinki.fi)

PS. Yes - it's free of any minix code, and it has a multi-threaded fs. It is NOT portable (uses 386 task switching etc), and it probably never will support anything other than AT-harddisks, as that's all I have :-).

He wrote the first Linux kernel in 1991. Two years after Linus' post, there were 12,000 Linux users. Linux is an essentially full UNIX clone. And now the Linux kernel is over six million lines of code! The main reason why Linux gained its popularity is because it was released under the GNU GPL licence (arguably one of the strongest 'copyleft' or quid pro quo licences). Later, Slackware, Red Hat, SuSE, Debian, et al, sprung up to provide packaged